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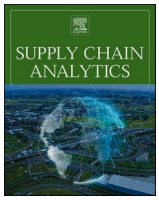

A bibliometric analysis of data-driven technologies in digital supply chains

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ABSTRACT

Internet of Things (IoT) and Cyber-Physical Systems (CPS) are the core components of data-driven technologies of Industry 4.0, attracting much attention in digital supply chains and leading to a growing tide of academic publications. This study conducts a bibliometric analysis of data-driven technologies in digital supply chains. Additionally, some bibliometric methods, such as co-word analysis, are utilized to study the intellectual structure of the field and present a big picture. The co-word analysis maps data-driven technologies' intellectual structure in digital supply chains and logistics. 3887 publications from the Web of Science (WoS) and Scopus between 2010 and 2021 were collected and analyzed. Then, a strategic diagram is employed on the co-occurrence network, indicating each theme's current situation from two aspects of applicability and theory development. The study reveals that IoT and CPS technologies are in their infancy in digital supply chains and logistics, and additional studies are needed to fill the research gaps in this field.

1. Introduction

In the era of Industry 4.0, traditional supply chains are undergoing a transformative shift to digital supply chains (DSCs), capturing the interest of researchers and industry leaders alike [34,103]. Digital Supply Chain (DSC) technologies are categorized into three pivotal types: data-driven, knowledge-based, and decision-oriented. Data-driven technologies are responsible for the collection, storage, and exchange of data across various entities. Knowledge-based technologies delve into this data to distill valuable insights and knowledge. Finally, decision-oriented technologies leverage this accrued knowledge to empower managers in making informed and strategic decisions [146,17]. Data-driven technologies form the foundational layer upon which knowledge-based and decision-oriented technologies operate; thus, without data-driven technology, the other two cannot work efficiently. Thus, we concentrate on the data-driven technology aspect of DSCs in the current research.

Internet of Things (IoT) and Cyber-Physical Systems (CPS) are among the most critical data-driven technologies in DSCs [17]. These technologies are considered by many researchers to be the driving force behind Industry 4.0, fueling its growth and innovation [124,42]. CPSs enable physical environments to interact with humans by integrating computational and physical abilities [14]. The employment of emerging IoT technologies has opened up novel opportunities for CPSs, enabling the connection of cyber-physical devices via internet-based platforms. Integrating IoT and CPS shifts the focus of internet applications from mere digital data transformation to the active control of physical environments through cyber-physical devices [99,37]. From a network perspective, IoT is a set of heterogeneous and unique addressable cyber-physical objects such as Radio Frequency Identification (RFID), sensors, and smart devices connected through a worldwide network in a standard communication protocol [15]. The expansion of communication and control through CPS over the internet is known as the IoT [23]. Indeed, these two concepts are interrelated and cannot be considered in isolation. Those publications that only focused on one of these technologies to review DSCs cannot provide a comprehensive overview of the data-driven dimension. The current paper is one of the few studies that address this gap by examining IoT and CPS concurrently in the context of DSCs. Fig. 1 presents a schematic representation of how these leading technologies are integrated within DSCs.

Both IoT and CPS are recognized as underlying technologies that facilitate compatibility among new technological developments, innovative ideas, and frameworks. They also lay the groundwork for ongoing research that propels the domain's continued progression [99]. In the

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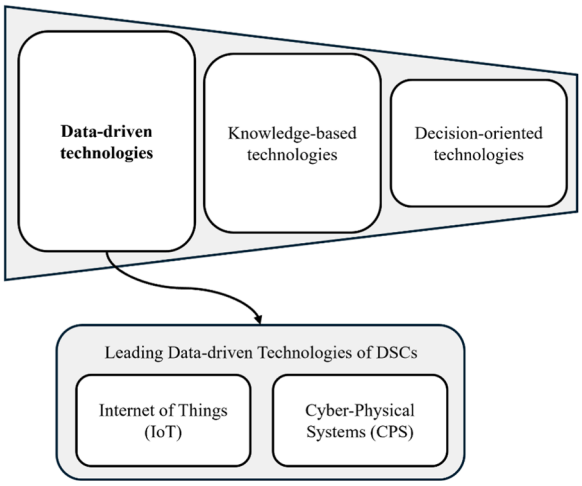

Fig. 1. Different technologies of DSCs.

context of digital supply chains and Industry 4.0, underlying technologies could include the hardware, software, networks, and protocols that form the backbone of IoT and CPS, allowing them to operate and communicate effectively. Therefore, by concentrating our research on IoT and CPS, we expect our study to cover various related data-driven technologies within DSCs. While there are various data-driven technologies, they do not serve as foundational elements in DSCs to the same extent as IoT and CPS. Including these other technologies in an analysis could skew the results, as they are not as essential to the core structure and functioning of DSCs as IoT and CPS.

The increasing volume of research on IoT and CPS technologies underscores the need for both scholars and practitioners to have a broad-scale understanding of these areas. Due to the insufficient capability of review analysis approaches to cover a wide range of publications, there is no comprehensive overview of data-driven technologies within the archipelago of conducted reviews. In contrast, bibliometric analysis serves as a suitable alternative, offering the capacity to effectively analyze an extensive corpus of publications [66]. It can be concluded that bibliometric analysis excels in managing extensive datasets, offering a macro-level perspective on a research domain. Conversely, review analysis delivers an in-depth and meticulous consolidation of findings tailored to a specific narrow subject matter (e.g., Vallaster et al. [134]). The advocate for bibliometric analysis emphasizes its capacity for autonomously producing research themes, which are likely to be less subjective than those derived through manual compilation in systematic literature reviews [80]. Bibliometric analysis serves as a powerful tool for researchers to uncover the intricate structure and evolutionary trends within a field, as noted by Pourhatami et al. [114].

While bibliometric methodologies have been applied to scrutinize sectors such as IoT and CPS, the focus on supply chains and logistics remained scant (e.g., [30,53,107,122,128,143]). Baziyad et al. [17] bridged this void by probing into IoT and CPS within DSCs through a literature review analysis. Nevertheless, the field still awaits the completion of a thorough bibliometric analysis. Building upon this premise, the present paper employs bibliometric analysis to unravel the synergy between the vanguard technologies of Industry 4.0—specifically, IoT and CPS—and the digital supply chain sphere. Such an analysis reveals practical implications for implementing IoT and CPS in DSCs, offering businesses a roadmap for integration and adoption. It also contributes to the theoretical understanding of digital supply chains, providing a foundation for future research and development in this area.

Although a few researchers have tried to employ bibliometric analysis for IoT in DSCs, they have ignored the essential role of CPSs [118, 30]. The current paper represents a novel bibliometric analysis that examines the IoT and CPS technologies, offering a novel perspective on the influence of data-driven technologies within DSCs. Previous studies have revealed various themes within the research field but have not examined the developmental progress of each theme. Indeed, the themes should be explained, how they can be applied in real applications, and how they can be developed. As a result, this paper employed a strategic diagram for analyzing the situations of different themes of IoT and CPS technologies in the context of DSCs, from the viewpoints of applicability and development.

Previous studies have often examined IoT and CPS in isolation, failing to provide a comprehensive view of their combined role in DSCs. This research addresses this gap by concurrently analyzing both IoT and CPS within the context of DSCs. We pioneer bibliometric analysis in DSCs, focusing on CPS and the IoT. It presents a strategic diagram to evaluate IoT and CPS's current state and future potential in DSCs. This paper offers an exhaustive analysis of the roles of IoT and CPS in DSCs by examining a vast array of papers from Web of Science (WoS) and Elsevier's Scopus databases, spanning 2010–2021, using co-word analysis. This technique unveils the latent knowledge within the field, allowing for a detailed understanding of the current research landscape and future trajectories. The paper aims to extract key insights from these core data-driven technologies and poses relevant research questions as follows: (a) What are the principal research themes and the latest advancements within each domain that are situated at the intersection of fundamental data-centric technologies in digital supply chains? (b) How has research on the topic evolved? (c) What are the specific enabling technologies that have been incorporated, and in what ways have they been merged with the recognized themes?

Overall, the current paper distinguishes itself with several key contributions to the field: (a) This study is one of the early efforts to investigate IoT and CPS technologies as fundamental elements of data-driven Industry 4.0 technologies, specifically within the realm of digital supply chains; (b) By focusing on IoT and CPS, the study aims to encompass a broad spectrum of related data-driven technologies within DSCs, recognizing their pivotal role compared to other technologies; (c) It introduces a strategic diagram to assess the applicability and development of various themes related to IoT and CPS technologies in DSCs; (d) Diverging from the traditional, time-intensive approach of manually standardizing keywords, this manuscript introduces a groundbreaking framework that utilizes string similarity, streamlining the keyword standardization process for extensive datasets. This paper's remainder is structured as follows: Section 2 provides a literature review of bibliometric and co-word analysis. Section 3 introduces the utilized methodology. Section 4 depicts the obtained results, and Section 5 explains the conclusion, limitations, and future directions.

2. Leading data-driven technologies of DSCs

Industry 4.0, representing a holistic approach to digitalization, stands at the forefront of technological exploration conducted through bibliometric analysis in the realm of digital supply chains. For instance, Motallebi et al., [108] provided an extensive survey of the architecture, progression, and movement of Industry 4.0 within supply chain research, utilizing analysis of academic social networks. Majiwala and Kant [104] conducted an exhaustive bibliometric analysis, encompassing 1554 articles from the Scopus database, spanning the years 2011–2022. They aimed to pinpoint research hotspots, thematic areas, trending topics, and the foundational knowledge base of Industry 4.0 as it intertwines with supply chain management. Hoang et al. [77] presented a comprehensive overview and organizational structure of knowledge on social media's role in supply chain management. It employed bibliometric analysis of 354 articles from the Web of Science database, spanning 2008–2022, using co-citation and co-word analysis techniques. The study identified five key ways social media is applied in supply chains: (1) improving sustainability and business transformation, (2) utilizing social media analytics to derive business insights, (3)

enhancing communication and coordination within the network, (4) fostering social engagement and customer relations, and (5) advancing supply chain relationship management strategies. Jetty and Afshan [84] provided a comprehensive bibliometric analysis and a systematic review of the literature on Industry 4.0 within the supply chain sector, aiming to discern the current research trends. In certain instances, bibliometric researchers honed in on more specialized areas of study within digital supply chains. Gharaibeh et al., [56] assessed the advancement of digitalization within construction supply chains through a thematic review of literature and identification of research trajectories. It offered a conceptual framework that assists researchers in examining the prevailing trends of Supply Chain 4.0, particularly its implementation in the wood construction sector relative to other industries. They also proposed potential pathways for future investigations in the domain of Supply Chain 4.0 within the wood construction industry. As digital technologies gain prominence in the Industry 4.0 landscape, bibliometric researchers have increasingly concentrated on particular technologies such as, artificial intelligence [115], blockchain [69], digital twin [96]. In alignment with contemporary research objectives, this has led to a detailed examination of the IoT and CPS, which are pivotal in driving the digital transformation of supply chains.

2.1. Cyber-physical systems (CPSs)

CPS was introduced by Helen Gill in 2006 at the National Science Foundation in the United States [98]. All in all, CPSs refer to a set of transformative technologies increasing the interactions between physical properties and computation systems for managing the interconnected systems [14] that have penetrated various domains such as smart grids, smart cities, smart manufacturing, and intelligent transportation (D. [150]). CPSs are regarded as physical and engineered systems integrating computational and physical capabilities and enable humans to interact with physical environments by enabling technologies such as monitoring and controlling systems (e.g., [70,93]).

2.2. Internet of things (IoT)

IoT was introduced by Kevin Ashton in 1999 [9], comprising a set of interconnected objects equipped with intelligence characteristics everywhere and every day [141]. In other words, physical and virtual environments can be connected and communicated through IoT technologies facilitating information-sharing processes [11]. Also, every object in an IoT-based platform is addressable, traceable, and identifiable among some sensors collecting the environmental data [10] under a common infrastructure [62]. In a comprehensive definition, IoT is a collection of smart objects armed with integrated and collaborated forms of sensors, networking, and processing technologies delivering smart services to the ultimate customers [8]. IoT is a transformation system converting the physical environment into a virtual one to provide an appropriate environment for a device to device communication (e.g., [119,142]).

By using bibliometric analysis, Bouzembrak et al. [30] could map out the landscape of IoT technology in food safety, identifying key contributors, influential regions, and the main topics of interest within this specific area of study. This method provides a structured way to visualize the complex web of academic work and to understand the collective direction of research efforts in the field. Kaya et al. [87] utilized bibliometric analysis to delve into the impact of the IoT on supply chain management within the context of Industry 4.0. In a same way, Through bibliometric analysis, Katoch [86] were able to systematically map out the landscape of IoT in SCM and logistics, offering a structured overview of the field's evolution and current state. This method allowed for a data-driven approach to identify trends, gaps, and opportunities for further research and practical application. Kayvanfar et al. [88] conducted a thorough co-word analysis of web-based content to present an extensive overview of the contemporary landscape of IoT in supply chains and logistics, underscoring the pivotal influence of IoT technology in advancing supply chain processes.

2.3. Digital supply chains (DSCs) in the era of leading data-driven technologies

While CPS and IoT are closely related fields that cannot be separated clearly, many researchers have not considered them together in their review research. Several research papers addressed IoT technologies in supply chains ([125,21,72]; Yangke [47]), while others emphasized CPSs [133,82]. The review of digital supply chains rarely considered both CPS and IoT technologies [105]. Consequently, considering IoT and CPS technologies in DSCs remains challenging, limiting researchers' and practitioners' access to an overview of these two leading technologies.

Using bibliometric analysis, Bouzembrak et al., [30] analyzed 48 academic publications which covered IoT applications in food safety. It cannot cover different supply chains, as it is limited to the food sector. In order to extend the analytical domain, Rejeb et al., [118] applied bibliometric analysis to a broader range of publications (807 papers) for investigating IoT technologies in supply chains. However, there is still an issue. Aside from IoT, neither study considered CPS. In Table 1, related works are briefly summarized.

While some studies have examined the integration of IoT and CPS within supply chain queries, the inclusion of an extensive array of non-relevant keywords has diluted their focus, diverting it from a strictly data-driven approach (e.g., [80,104]). For example, Iftikhar et al. [80] expanded their search criteria to include not only IoT and CPS but also blockchain, big data, cloud computing, and other cutting-edge technologies to gather comprehensive data for their analysis. This issue is also prevalent in the expansive domain analyses such as Industry 4.0 within supply chains, as outlined at the start of Section 2. The current study is at the forefront of bibliometric research, addressing a critical gap by concentrating exclusively on IoT and CPS within supply chains—a subject that, until now, has not been thoroughly explored. By pioneering this focused approach, our analysis offers unprecedented insights into the integration and influence of these technologies in DSCs. The findings have the potential to shape future research directions and inform industry practices, marking a significant advancement in the field.

The present work focuses solely on revealing the various clusters of investigated publications, without analyzing their relative status. The most popular methodology for analyzing cluster status is the strategic diagram, which is not present in papers introduced by Bouzembrak et al., [30] and [118].Using a strategic diagram, we can measure how each field is being developed, as well as how it can be utilized in various applications. Besides showing the different clusters of IoT and CPS within DSCs, the status of each cluster is also determined by a strategic diagram.

3. Materials and methods

Bibliometric analysis employs mathematical and statistical techniques to scrutinize written communication and literature. Essentially, it is a quantitative approach that allows for the identification of prevailing trends within a research field [43]. Among the innovative tools in bibliometrics, text mining stands out for its ability to unearth the latent knowledge within scholarly articles [147]. Furthermore, co-word analysis, a subset of text mining [97], has been effectively used in the literature for bibliometric studies [135] to delineate the intellectual contours across various disciplines [114,20]. Co-word analysis is a quantitative technique that reveals the connections and interplay among concepts within a field (Q.-R. [149]). As a graph-based method, co-word analysis focuses on dividing a co-word network into sub-themes and sub-graphs, thereby mapping out the diverse sectors within the domain under study [40]. The synergy between bibliometric and co-word analyses is depicted in Fig. 2, underscoring their role as pivotal tools in

Table 1
Brief overview of related works in comparison to the present study.

References	Area				Methodology	
	CPSs	IoT	Logistics	Supply Chains	Review	Bibliometric
Ivanov, Sokolov [82]	√			√	√	
Tonelli et al., [133]	√			√	√	
Li [100]		√		√	√	
Sun [131]		√	√			
Verdouw et al., [136]		√	√	√	√	
Manavalan, Jayakrishna [105]	√	√	√	√	√	
Aryal et al., [7]		√		√	√	
Birkel, Hartmann [26]		√		√	√	
Ben-Daya et al., [21]		√	√	√	√	
Rejeb et al., [117]		√	√	√	√	
Shah, Ververi [125]		√		√	√	
He et al., [72]		√		√	√	
Kaya et al., [87]		√		√		√
Ben-Daya et al., [22]		√	√			
Ding et al., [47]		√	√			
Baziyad et al., [17]	√	√		√	√	
Bouzembrak et al., [30]		√		√		√
Rejeb et al., [118]		√	√	√		√
Katoch [86]		√		√		√
Abbasi, Ahmadi Choukolaei [1]				√	√	
Bouchenine, Abdel-Aal [28]				√		√
Emrouznejad et al., [49]				√		√
Aljuneidi et al., [4]				√	√	
Abosuliman [2]	√			√	√	
Canonico, Sperlì [36]	√				√	
Singh et al., [128]	√	√				√
Kayvanfar et al., [88]		√	√	√	√	
Alvarez-Alvarado et al., [5]	√				√	
Hasan et al., [71]	√				√	
Our Paper	√	√	√	√		√

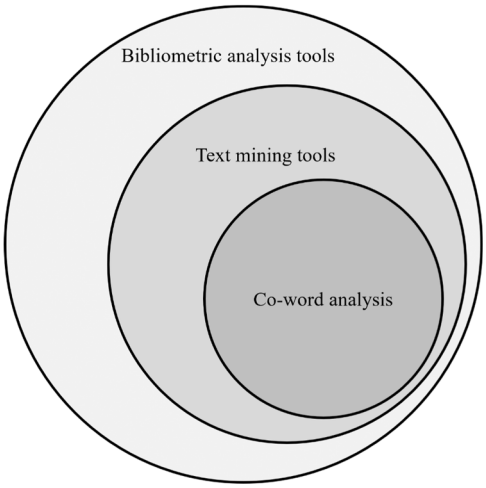

Fig. 2. The position of co-word analysis and text mining tools in bibliometric analysis.

bibliometric research.

Overall, co-word analysis is particularly valuable in fields with large volumes of literature, as it can efficiently process and analyze extensive datasets to provide a macro-level understanding of the research landscape. It's a powerful tool for researchers, policymakers, and institutions aiming to grasp the complexities and dynamics of scientific knowledge. Thus, this research uses a co-word analysis to disclose the hidden knowledge of IoT and CPS-based supply chain publications. Different themes and sub-themes of the surveyed field are revealed as the first extracted knowledge. Finally, themes are analyzed from applicability and theory development perspectives by applying a strategic diagram. Hosseini et al., [78] introduced a comprehensive framework for co-word analysis applied in this paper. According to their framework, a co-word analysis comprises four main phases; data collection, keyword selection, matrix calculation, and results.

3.1. Data collection

WoS and Scopus are two central databases that researchers mainly refer to and have enough coverage of interdisciplinary publications (Archambault et al., 2006). After the combination of the search queries extracted from the literature review (see Table 1), the main query used in this paper is TITLE-ABS-KEY ("cyber-physical system*" OR "cyber physical system*" OR "smart system*" OR "cyberphysical system*" OR "CPS*" OR "Internet of Things" OR "Internet-of-Things" OR "IoT" OR "Internet of Everything*" OR "Industrial Internet" OR "Web of things" OR "Web-of-Things" OR "WoT") AND ("supply chain*" OR "logistics" OR "Supply Chain Management" OR "Logistics Management" OR "Digital Supply Chain*" OR "digital logistics" OR "Smart Supply Chain*" OR "Intelligent Supply Chain*" OR "Smart Logistics" OR "Intelligent Logistics"). The search query, spanning from 2010 to March 2021, targeted titles, keywords, and abstracts of various scholarly works, including journal articles, conference papers, reviews, book chapters, and editorials. Only papers published in English were included. This could be due to the researchers' language proficiency or the wider accessibility of English-language research. The search targeted a variety of scholarly documents, including journal articles, conference papers, reviews, book chapters, and editorials. This broad scope allows for a comprehensive understanding of the field. Publications that were irrelevant or off-topic were excluded. This means that only papers directly related to the research question or topic were considered. Publication slacking keywords and abstracts were omitted. Keywords and abstracts are essential for understanding the focus and content of a paper without having to read the full text.

The initial search yielded 2070 publications from the Web of Science (WoS) and 3961 from Scopus. After deduplication, the total was consolidated to 4514 entries. Subsequent filtering removed 308 publications without keywords and seven lacking abstracts. An additional

312 publications were deemed irrelevant and excluded. Fig. 3 details the data refinement process, and Fig. 4 highlights the marked growth in publication volume between 2010 and 2021, emphasizing a pronounced escalation commencing in 2016.

3.2. Keyword selection

Selecting the right keywords is pivotal in co-word analysis, as it has a direct impact on the quality of the results. By choosing the most pertinent keywords for constructing the co-word network, one can extract more meaningful and in-depth insights from the co-occurrence network. Frequency is a key criterion in this selection process, with an emphasis on identifying and incorporating the most commonly occurring keywords into the co-word graph [78]. After extracting the primary list of keywords, a filtering process is needed to integrate and clean the collected data. Keyword filtering comprises three main phases, i.e., a) standardizing the similar keywords (e.g., *supply chain* and *supply chains* or, e.g., *radio frequency identification* and *radio-frequency identification*); b) integrating acrimonies (e.g., *Internet of Things* and *IoT*); c) removing general keywords (e.g., *algorithms*).

3.2.1. Standardizing the similar keywords

While previous research often relied on manual standardization of keywords—a laborious task with large datasets—this paper presents an innovative framework that leverages string similarity for efficient keyword standardization. We employ six key similarity metrics: Hamming [68], Levenshtein [60], Jaro Winkler [140], Jaccard [83], Sorensen [46], and Ratcliff Obershelp [116] facilitated by the 'textdistance 4.2.1' package in Python 3.6. This approach ensures that keywords with high similarity scores, such as 'supply chain' and 'supply chains,' are accurately consolidated. The entire standardization process is detailed in Fig. 5.

3.2.2. Integrating acrimonies

After consolidating similar keywords, acronyms were identified and substituted with their full terms to enhance clarity (for example, *'SCM'* was expanded to *'supply chain management'*). Additionally, for the sake of graph legibility, acronyms were employed to represent lengthy terms.

3.2.3. Removing general keywords

In the context of a co-word network, general keywords are those that are common and may not provide specific insights into a particular research domain. They are often too broad to be of interest to domain experts and may not discriminate between the nuanced topics of a field.

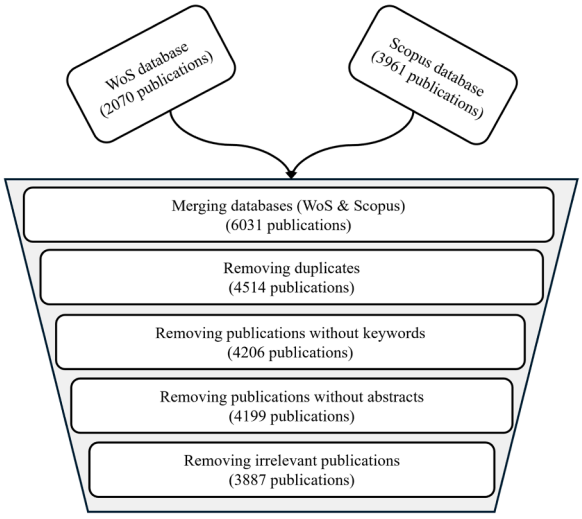

Fig. 3. Search query results from WoS and Scopus databases.

For example, words like "data," "analysis," or "results" could be considered general because they are widely used across various research areas and do not pinpoint a unique aspect of a study. Consequently, we exclude general keywords to refine the analysis process.

3.3. Matrix calculation

The co-occurrence matrix is fundamental to co-word analysis, as it records the frequency of keyword pairings. Before constructing the co-word network, it is essential to normalize the matrix. Various normalization methods have been adopted in literature, including the Jaccard Index, Association Strength, Cosine, and Inclusion Index [48], which are detailed in Table 2 and in terms of Eqs. (1−4). Association Strength, which excels in modularity—a key measure of community detection quality—outperforms other normalization methods. Consequently, it is the preferred choice for normalizing the co-word network.

Where C_{ij} represents the number of co-occurrences of keywords i and j. Besides, C_i and C_j indicate the frequency of keywords i and j, respectively. According to the normalized co-occurrence matrix, the co-word network can be visualized in which each node is a keyword, and edges are the normalized co-occurrences.

3.4. Content analysis

In order to find the involved themes and topics of the research field, clustering and community detection methods are applied on normalized co-word matrices [19,78]. Also, a strategic diagram is utilized to evaluate and compare the positioning of each identified theme relative to others. The integration of co-word analysis with strategic diagrams enriches the bibliometric analysis by providing a multi-dimensional perspective that is more informative and actionable for understanding and navigating the complexities of scientific research.

3.4.1. Themes identification

Community detection algorithms are a type of clustering method concentering on graph-based datasets such as co-occurrence networks. The Louvain algorithm [27] is one of the most used community detection algorithms applied in this paper for clustering the co-word network among Gephi 0.9.2 software. Also, community detection is evaluated by the modularity criterion.

Modularity is an appropriate measurement for discovering community detection algorithms. More modularity values indicate that the provided themes have the lowest similarity with others and items in a community are very similar [110]. The modularity formula is presented in Eq. (5) [110].

$$Q = \frac{1}{4}\sum_{ij}\left(A_{ij} - \frac{k_i k_j}{2m}\right)s_i s_j \tag{5}$$

$$m = \frac{1}{2}\sum_{i} k_i$$

Where k_i and k_j represent the node's degree and m indicates a total number of edges. A_{ij} introduces presented edges between node i and j. Also, the expected number of edges between nodes i and j can be calculated as $\frac{k_i k_j}{2m}$.

3.4.2. Strategic diagram

A strategic diagram maps science clusters in a two-dimensional matrix, determining their maturity [90]. A strategic diagram comprises two main measurements, i.e., density and centrality (or degree centrality). Density is an appropriate criterion for measuring a theme's internal cohesion, and centrality measurement represent how each theme can play a central role in the research field. According to Hosseini et al., [78], the metrics of density and centrality are determined using Eq. (6) and Eq. (7). The strategic diagram, detailed in Fig. 6, is divided

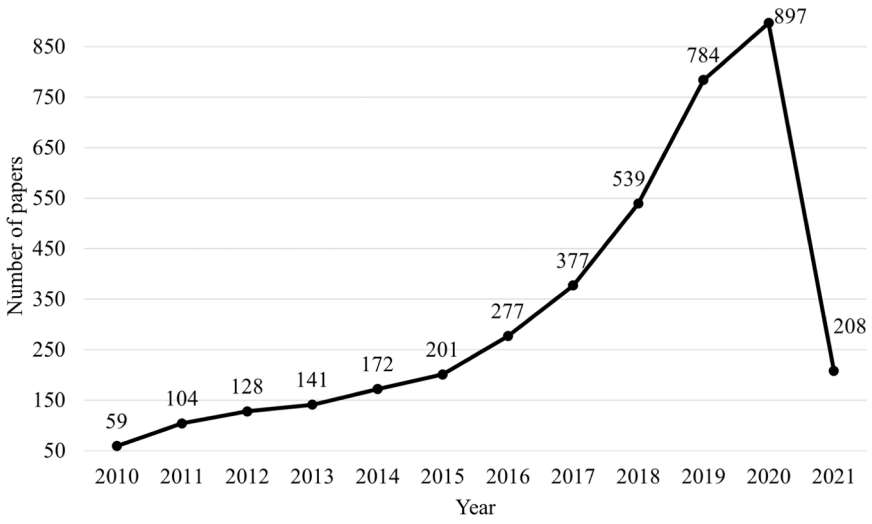

Fig. 4. Number of publications between 2010 and 2021 (IoT and CPS- based supply chains).

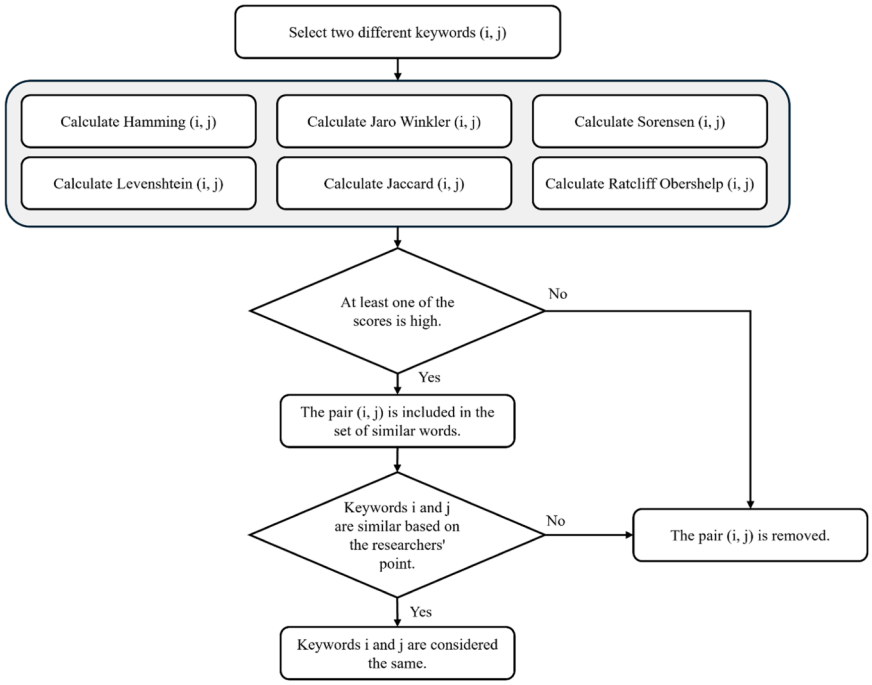

Fig. 5. Standardizing similar keywords

Table 2
Search queries of IoT, CPSs, and supply chains from the literature.

Domain	Related Search Query	Paper
CPSs	"cyber-physical system*" OR "cyber physical system*" OR "smart system*" OR "cyberphysical system" OR "CPS"	Mohamed et al., [107]
IoT	Internet of Things OR Internet-of-Things OR IoT OR Internet of Everything OR Industrial Internet OR Web of things OR Web-of-Things OR WoT	Koot et al., [94]
Supply chains	*"supply chain*"* OR "Supply Chain Management" OR "Logistics Management" OR	(Nakamura et al., n. d.; [50,118,94])

into four principal quadrants [58].

$$C_L = \sum_{i \in L} \sum_{j \in M} W_{ij}.e_{ij} \tag{6}$$

In Eq. (6), C_L is known as the centrality of community (theme) L, and i indicates the nodes (words) of community L. M refers to other themes except community L, and j shows the existed nodes of these communities. e_{ij} is a binary variable indicating the existence or non-existence of a link between nodes i and j. Finally, W_{ij} refers to the weight of the existed link between nodes i and j.

$$D_L = \frac{2E}{N(N-1)} \tag{7}$$

In Eq. (7), D_L indicates the density of cluster L, E represents the total number of edges in theme L, and finally, N is the total number of nodes in cluster L. The strategic map is divided into four main sections, including *Mainstreams* (quadrant I), *Ivory Tower* (quadrant II), *Bandwagon*

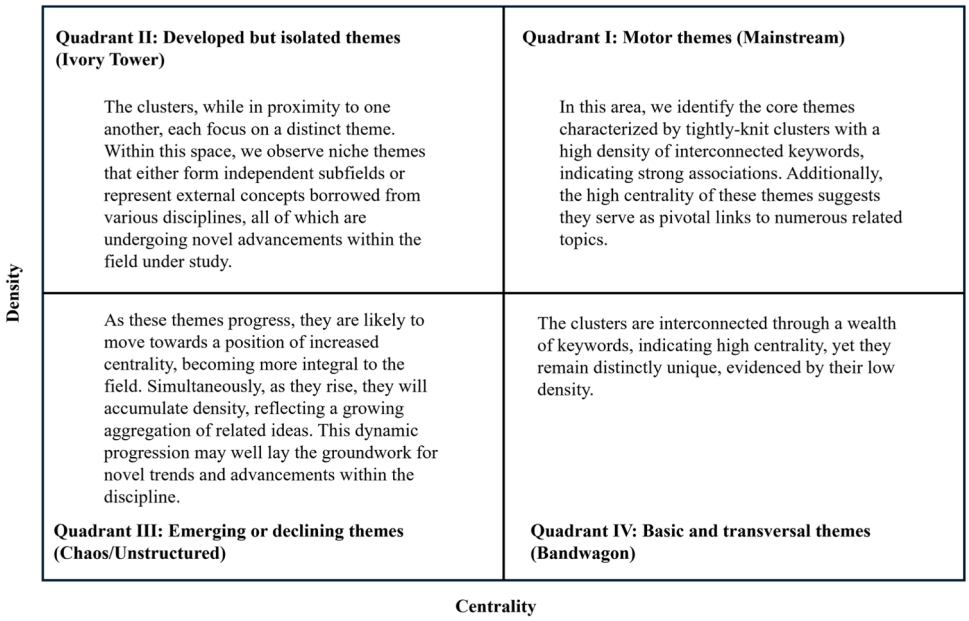

Fig. 6. Strategic diagram based on density and centrality measurements (adapted from Giannakos et al. [58].

(quadrant III), and *Chaos/Unstructured* (quadrant IV). Emerging or declining themes are not developed enough and cannot be combined with other areas resulting in the undeveloped and peripheral field. Over time, the situation of an emerging or declining theme can be changed in two ways:

a) *Further development and move towards Ivory Tower:* When an emerging or declining theme is developed, it moves to the Ivory Tower quadrant. Although a field has developed in the ivory tower section, it is still in an isolated environment that does not connect enough with other themes. After moving towards the ivory tower, connecting to other fields, and finding more theme applications, it shifts to the mainstream theme. Themes in the mainstream quadrant have developed enough and linked intensely to other fields, also known as 'motor themes'.
b) *Further cohesion and move towards Bandwagon:* When an emerging or declining theme links to other themes and shows its applications in other concepts, it gains more centrality and moves to the bandwagon theme. Bandwagon theme includes basic and transversal themes having low density but high centrality. In other words, despite being a beginner have found many fans in a wide range of applications and domains. Like the ivory tower quadrant, bandwagon themes can be shifted to the mainstream theme, but with improvements in their density, no centrality.

4. Result

4.1. Different areas of CPS & IoT-based supply chains

The Louvain community detection algorithm was applied for theme identification of CPS & IoT-based supply chains domain leading to disclose nine main themes: ''project management'', ''Enabling Technologies'', ''Location-based Services (LBSs)'', ''Agri-Food Supply Chains'', ''Smart Grid Challenges'', ''Industry 4.0'', ''Intelligent Computing'', ''Marketing, and ''Data Science''. The co-word network, depicted in Fig. 7, is visually organized by color-coding, with each distinct color denoting a separate theme. In the co-word network visualization (Fig. 7), keywords that appear more frequently are displayed in larger fonts for emphasis. For detailed analysis, community detection is conducted on sub-themes comprising over 40 nodes. The preliminary findings of this re-community detection, including the number of nodes, edges, and sub-themes for each primary theme, are summarized in Table 3 and illustrated in Fig. 8. Also, the modularity criterion is calculated for each clustering. Each theme of the main co-word network is explained (Subsections 4.2.1 to 4.2.9). According to several studies, modularity should be greater than 0.2 in a successful community detection (e.g., [55,3,127]). After applying the Louvain algorithm to the main graph and its sub-graphs, we found that all modularity criteria exceeded the threshold of 0.2, as illustrated in Table 3. Hence, we can say that the discovered communities are of high quality. In other words, clusters' internal members are similar and different from the members of other clusters.

Overall, the main themes and sub-themes of CPS and IoT-based supply chains and logistics domains over the last decade are depicted in Fig. 7.

4.1.1. Project management (C1)

The successful deployment of Industry 4.0 technologies within supply chains necessitates sophisticated skills and methodologies. A robust project management framework is essential to effectively orchestrate these competencies. Despite its critical role, project management has not been adequately emphasized in the context of Industry 4.0 and, by extension, Supply Chain 4.0—which leverages these advanced technologies. Recognizing project management as a fundamental element is vital for the effective integration of technologies like IoT and CPS into supply chain operations [52]. Prior to the adoption of IoT and CPS technologies in supply chains, it is imperative to address several foundational prerequisites during the project management phase. Key project management tasks in the development of IoT and CPS-enabled supply chains include designing technological architectures, standardizing supply chain operations, simulating process workflows, and conducting statistical analysis of the simulated data.

Computer simulations simulate different standards and architectures during the project management phase to implement IoT and CPS technologies in supply chains. Simulations are run, then the results are analyzed statistically, and then the best standards and architectures are selected. Fig. 9 shows an overview of project management tasks for implementing IoT and CPS technologies in supply chains.

It is important to remember that the project management process does not end with the mentioned tasks, and the provided figure is based on the analyzed literature. The project management community with 11 nodes and 21 edges can be seen in Fig. 10.

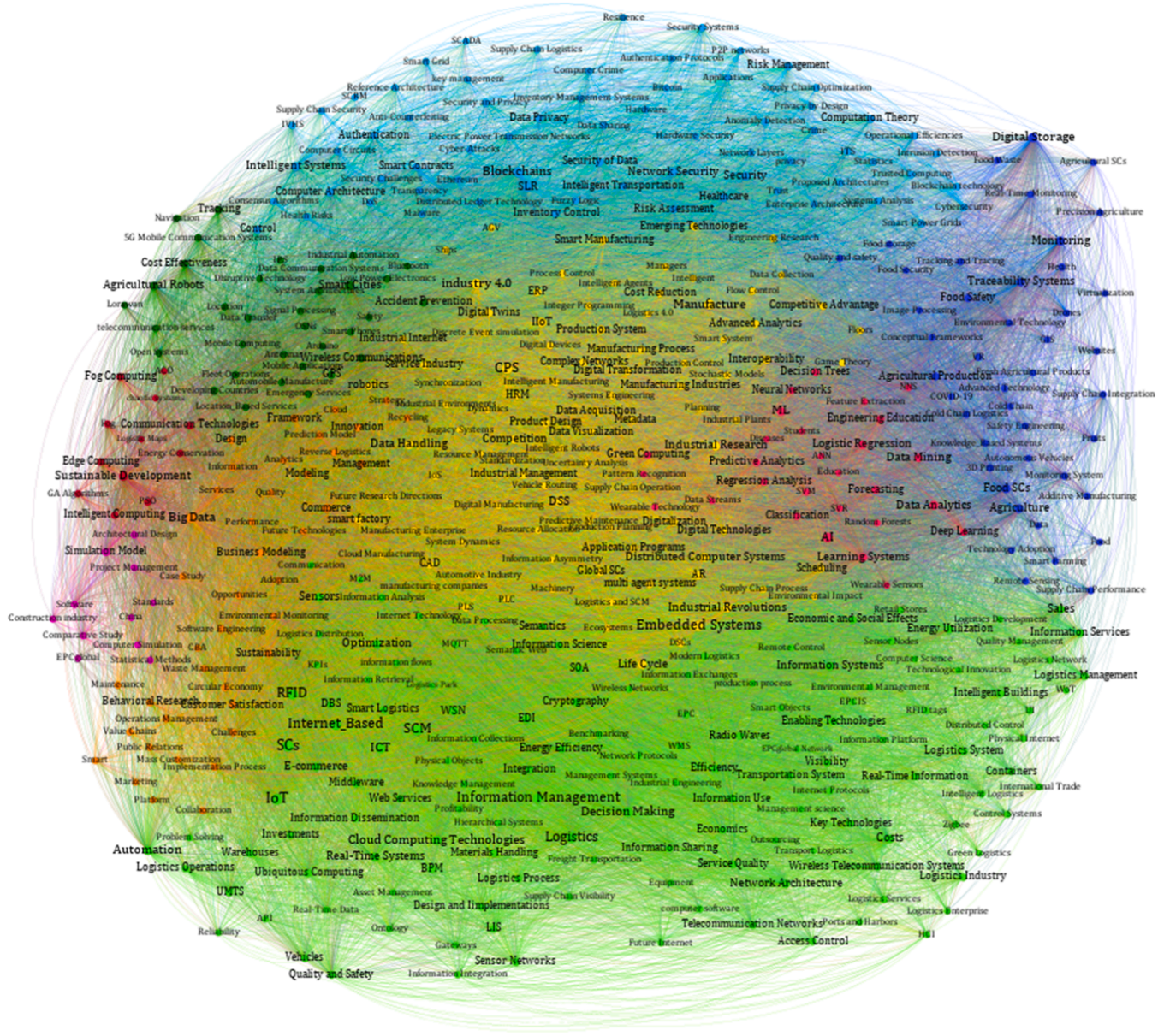

Fig. 7. The co-word network of CPS & IoT-based supply chains.

Table 3
Co-word normalization methods.

Normalization	Modularity	Eq.
$Association\ Strength(i,j) = \frac{C_{ij}}{C_i \times C_j}$	0.252	1
$Cosin(i,j) = \frac{C_{ij}}{\sqrt{C_i \times C_j}}$	0.205	2
$Jaccard(i,j) = \frac{C_{ij}}{C_i + C_j - C_{ij}}$	0.212	3
$Inclusion\ Index(i,j) = \frac{C_{ij}}{\min(C_i,\ C_j)}$	0.175	*4*

4.1.2. *Enabling technologies (C2)*

Many technologies, such as IoT and CPS, are not standalone technologies. However, instead, they require other technologies or enabling technologies to work. In other words, enabling technologies facilitate efficient utilization of main technologies. C2, with 138 nodes and 3803 edges, primarily focused on enabling technologies. After executing the Louvain algorithm on this theme, seven sub-themes are disclosed, as shown in Fig. 11.

4.1.2.1. Information management (C2_1): Blue Sub-theme. In information management, the flow of information is controlled, from how it is created, gathered, organized, and disseminated to how it is used. People and organizations use information management to access, process, and utilize information efficiently and effectively [44]. Based on the reviewed papers and prepared graph in Fig. 11, it can be found that information management tasks (data creation, collection, organization, dissemination, utilization) of IoT and CPS-based supply chains can be handled by different enabling technologies. Through technologies such as RFID, data can be created and collected by sensors embedded in supply chains and then disseminated via Internet-based and cloud computing technologies for storing in a database. The data are fed into information systems for virtualizing supply chains so that supply chains can be visualized. Indeed, information systems enable supply chains to represent physical objects virtually. C2_1 focuses on information management in supply chains based on IoT and CPS, showing different information management tasks.

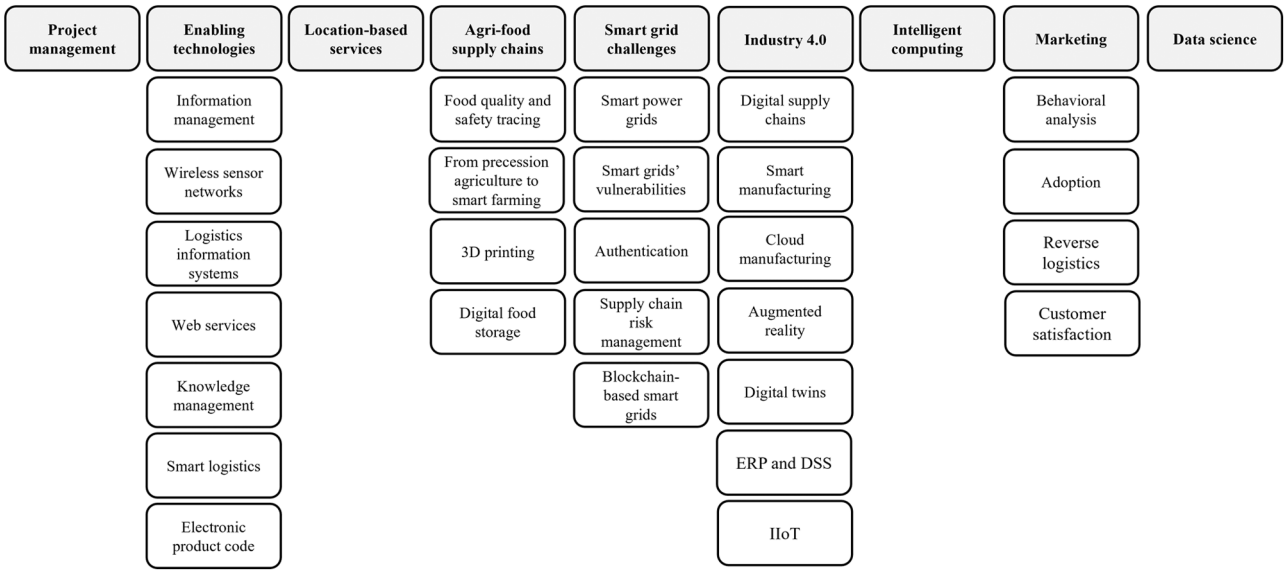

Fig. 8. The main themes of CPS and IoT-based supply chains and logistics.

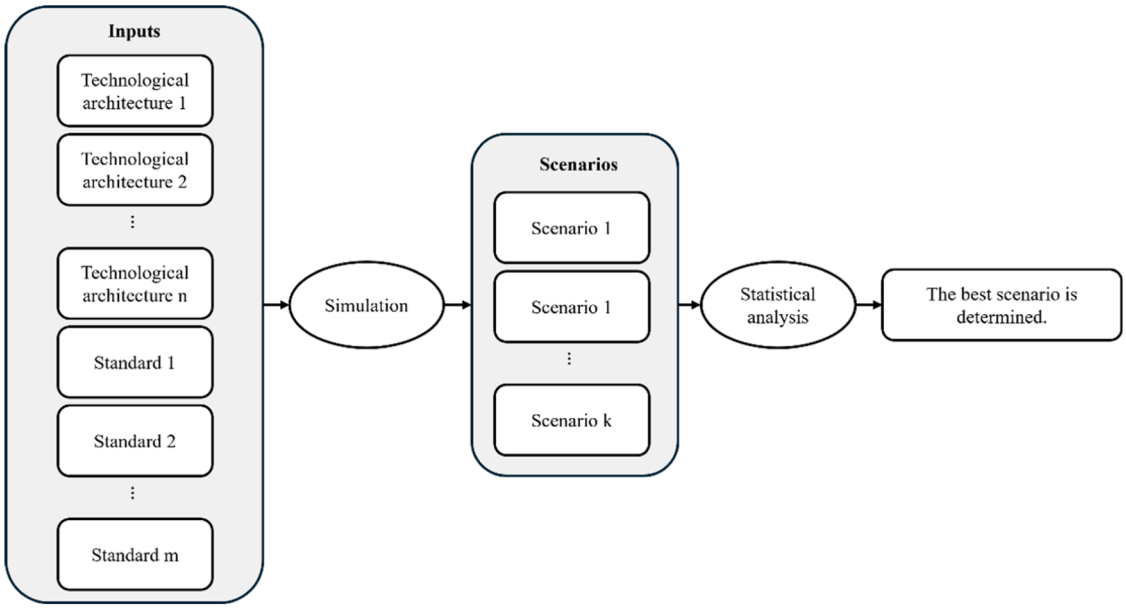

Fig. 9. Project management in the context of IoT and CPS implementation in supply chains.

4.1.2.2. Wireless sensor networks (WSNs) (C2_2): Green Sub-theme. Wireless Sensor Networks (WSN) are fundamental to the operation of IoT and CPS technologies. While WSNs can function independently of IoT and CPS, the latter technologies are inherently reliant on the principles of WSNs for their existence [45,89]. A WSN is a set of wireless sensors (nodes) which are interconnected to form a network and have three main capabilities: I) environmental data collection, II) data processing and III) wireless communication. In a WSN, data is collected from the environment and sent over wireless links to clients and commands. The received data can be computed for specific purposes [67]. In Fig. 11, it is evident that WSN technologies of IoT and CPS-based supply chains can be described based on sensation, communication, and computation perspectives.

The focus of the sensation is mainly on two concepts: physical objects and energy consumption. A physical object is something such as a vehicle or container in logistics and transportation systems that can be equipped with sensors. Since embedded sensor nodes have limited resources, energy efficiency has become a critical issue for SCs. In order to implement simple computation processes and aggregation functions (MEAN, MEDIAN, MAX, and MIN), the collected data from embedded sensors should be transferred to other nodes. Network protocols such as ZigBee and MQTT carry out transformation tasks. The sensor nodes of WSNs can handle basic analysis approaches, but due to the limited computational power, they cannot handle advanced analysis methods. To perform more sophisticated analyses, they connect to external networks such as Universal Mobile Telecommunication Systems (UMTSs) and other types of telecommunication networks via gateways.

4.1.2.3. Logistics information systems (LISs) (C2_3): Red Sub-theme. In summary, LIS is a supportive decision-making tool responsible for monitoring and controlling the supply chain's flows, including information, material, and financial ones. The main objective of LIS is to deliver the right amount of products at the right time and place [74,75]. C2_3 explains managing the modern logistics working based on Logistics

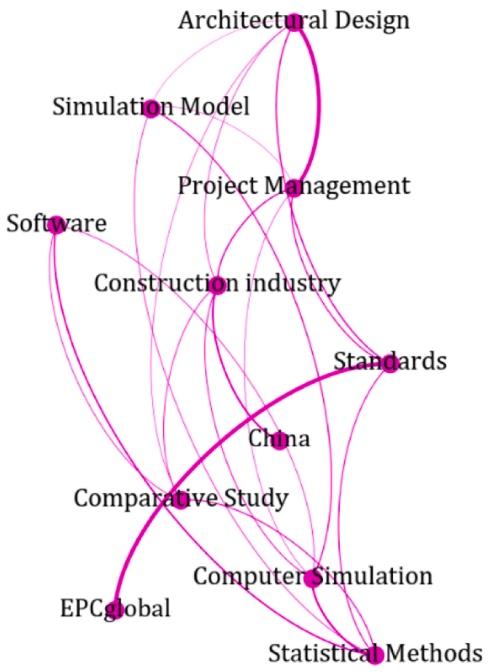

Fig. 10. Project management theme.

Information Systems (LISs). LIS-based logistics help to manage smart objects and materials among controlling the information flow. Indeed, LIS enables logistics industries to implement machine-to-machine (M2M) interactions via the internet, disclosing the intelligent logistics concept.

4.1.2.4. Web services (C2_4): Yellow Sub-theme. Web services are reusable software modules that connect customers with internet services and then provide related services for meeting customers' needs [112]. Using Service-Oriented Architectures (SOAs) as one of the main tools of web services, a software can be divided into multiple sub-applications that run on multiple remote servers, all of which can provide services to clients [85]. Even though SOA operates on a reactive approach, where clients' requests are processed and responded to, they are not proactive and cannot react to events. Thus, an Event-Driven Architecture (EDI) was developed to sense environmental events and respond to them in real-time [32]. Web services provide a wide range of services to end users but require more automation to handle tasks like search, choice, enhancement, and implementation of appropriate services. Adding machine-processable capabilities to web services is necessary for such automation. In this way, the semantic web is born [51]. The C2_4 illustrates the IoT and CPS technologies in the supply chain through the web service dimension. Web services in IoT and CPS-based supply chains have been studied from two perspectives: architecture and performance analysis. Architectures such as SOA and EDI relate to the structure of web services where the technological layers and how they interact to provide appropriate services are determined. In a more advanced mode, the semantic web employs ontology-based architectures to provide appropriate services. As part of the performance analysis of web services, key performance indicators (KPIs) such as reliability are studied to improve service provision.

4.1.2.5. Knowledge management (KM) (C2_5): Purple Sub-theme. KM consists of selecting, collecting, storing, organizing, packaging, and communicating the essential business information of a firm to enhance its competitive advantages by making the best decisions [24]. Based on the C2_5, it can be understood that IoT and CPS-based supply chains use knowledge management systems to analyze structured information of Warehouse Management Systems (WMSs) to extract relevant knowledge that may be useful in decision-making processes. Accordingly, KM aims to make IoT and CPS-based supply chains more efficient in environmental management, asset management, and quality and safety problem-solving.

4.1.2.6. Smart logistics (C2_6): Light-blue Sub-theme. Smart logistics harness Industry 4.0 technologies, notably the IoT and CPS, to foster integration of information, intelligence, and systematic operations within logistics distribution networks ([16]; N. [148,132]). Ports and harbors represent key areas of interest for the implementation of smart logistics, categorized within the C2_6 cluster. Presently, the Web of Things (WoT), serving as an application platform for IoT, is recognized as a pivotal technology in the realm of smart logistics.

4.1.2.7. Electronic Product Code (EPC) (C2_7): Orange Sub-theme. EPC is a tag data standard for identifying physical objects by employing a unique identifier developed by the Auto-ID Center at MIT. The EPCglobal Network is comprised of four parts, namely I) Object Name Service (ONS); II) EPC Information Services (EPCIS); III) EPC Discovery Service (EPCDS); IV) EPC Security Services (EPCSS) [31,81]. C2_7 refers to EPCIS, which enables logistics and supply chains to identify physical objects based on unique codes. The EPCIS is a crucial component of electronic tag-based technologies such as RFID to construct the EPC Network Architecture. An EPC Network Architecture as an information service provides a user interface that allows clients to trace and control data from different sources of EPC data.

4.1.3. Location-based services (LBSs) (C3)

The LBS mainly deals with location-based data gathered from (mobile) devices and users [79]. LBSs integrate location-based data from mobile devices and objects (e.g., users') information to offer value-added services for meeting users' needs over wireless networks [57]. Two main types of LBS exist user-requested and condition-based LBSs. The position data in a user-request scenario is retrieved once by the user and used for subsequent requests to location-dependent services. Navigation and routing are some of the critical user-requested LBS services. Condition-based LBSs, on the other hand, retrieve objects' positions once predefined conditions have been met. Emergency services can be considered condition-based LBSs, which automatically receive a person's location from the mobile network when they call the center [41]. Since different objects in a logistics system move between places, finding their position among supply chains is crucial. Various LBS services, such as location-based order allocation, navigation, and location-based pricing, are used today to improve logistics supply chains ([139]; Xu [138,76]).

As indicated by the dark green color in Fig. 7, C3, which has 34 vertices and 214 linkages, concentrated on the LBS concept of IoT and CPS technologies among logistics supply chains. In brief, C3 demonstrated four technology types of *location-aware technologies* (e.g., Global Positioning Systems (GPS), Indoor Positioning Systems (IPS), and Online Social Networks (OSNs)), *data collection technologies* (e.g., Arduino), *data transfer technologies* (e.g., Bluetooth, antennas, 5 G mobile communication systems, and wireless communication), *data processing technologies* (e.g., mobile applications, smartphones, mobile computation). LBSs combine object data collected from data technologies with location-based data gathered from location-aware technologies and transfer them using data transfer technologies to data processing technologies for service delivery (See Fig. 12). A variety of LBS applications in logistics supply chains have been developed in various domains, such as agricultural robots, fleet operations, emergency services, and accident prevention.

The related network of the C3 theme is illustrated in Fig. 13.

4.1.4. Agri-food supply chains (C4)

The significance of agriculture and food within supply chains is

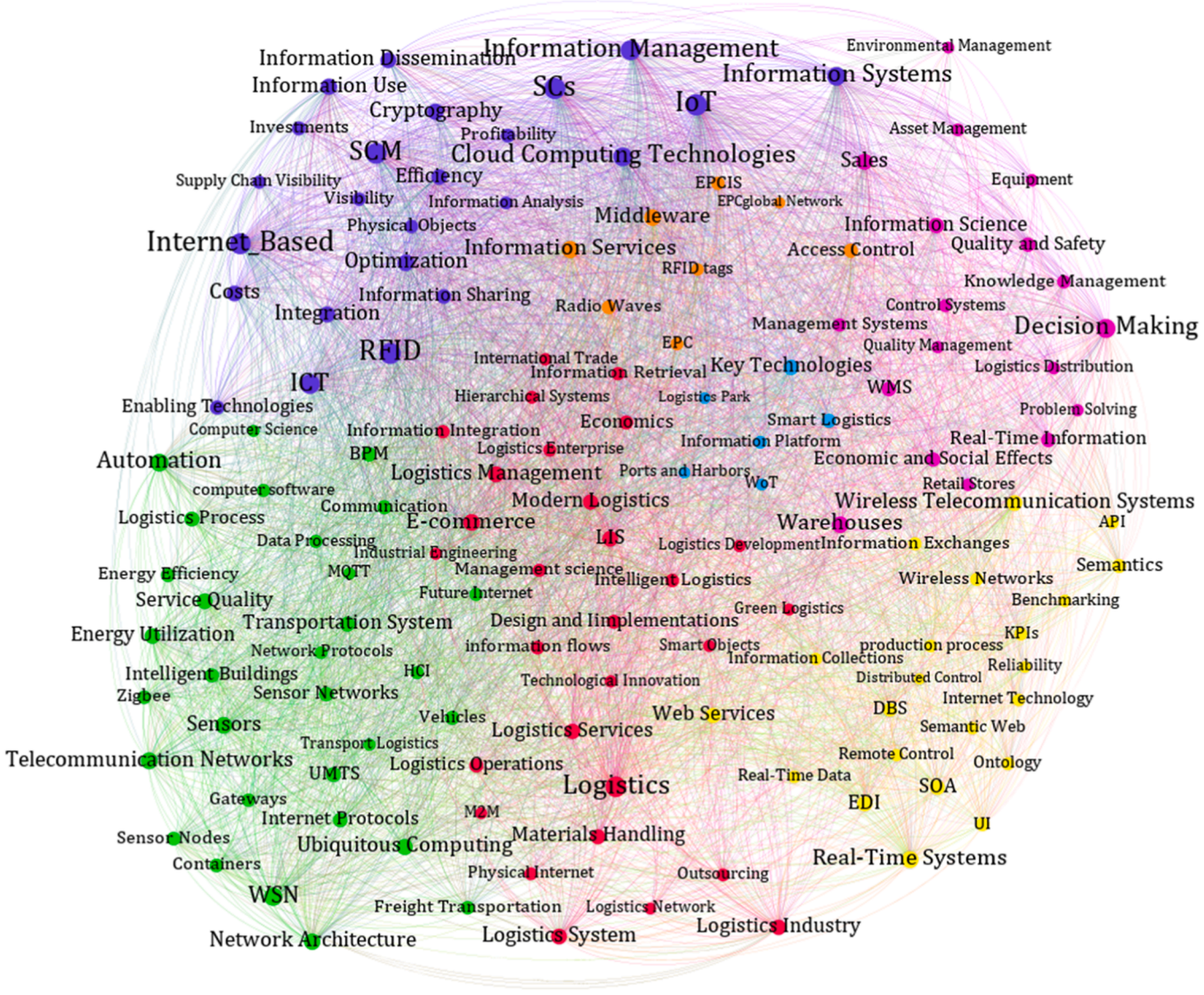

Fig. 11. Enabling technologies.

paramount, especially in today's global economy [126]. C4, with 43 nodes and 347 edges, is referred to as the Agri-food supply chain. After running the Louvain algorithm on this community, four sub-themes are revealed, as shown in Fig. 14.

4.1.4.1. Food quality and safety tracing (C4_1): Green Sub-theme. As a result of globalization, foods and agricultural products travel long distances from farm to fork. Therefore, quality assurance and safety assurance are critical issues for supply chains during travel. As part of Agri-Food supply chains, traceability is essential for meeting food safety and quality and increasing consumers' confidence [13]. C4_1 focuses on the tractability duty of Agri-food supply chains in which real-time monitoring systems trace fresh agricultural products' safety and quality.

4.1.4.2. From precession agriculture to smart farming(C4_2): Red Sub-theme. As a management strategy, precision agriculture (PA) utilizes information technologies for data collection from multiple sources, which can be applied as a decision support tool for crop-related productions. As a result of atomization processes, PA facilitates the collection and analysis of data, thereby allowing for quick decision-making and management process implementation [39]. With PA systems, data is stored in local databases without the possibility of sharing it with other farmers and suppliers. As emerging technologies such as the IoT emerged, PAs role faded, and a new concept called smart farming (SF) emerged. Food supply chains are equipped with IoT-based systems, which allow farmers to collect, share, and analyze data in real-time and make decisions faster [120]. However, some researchers consider both SF and PA to have the same meaning and use them interchangeably. Overall, SF and PA are defined as using modern technologies, such as IoT and drones, to manage farms to increase food production and yield while reducing the need for labs, land, and inputs [144]. C4_2 explains SF and PA concepts and their enabling technologies, such as Geographic Information Systems (GIS), knowledge-based systems, virtual reality (VR), environmental technologies, and drones.

4.1.4.3. 3D printing (C4_3): Yellow Sub-theme. Additive manufacturing (AM), also known as 3D printing, builds objects by joining materials based on 3D model data [35]. In other words, AM is a technology that can create physical models from virtual solid model data. In an AM technology, collections of 2D cross-sections with finite thicknesses are generated from the data. Physical objects are formed by adding 2D cross-sections together in a layer-by-layer sequence [59]. 3D printing plays a crucial role in food safety and quality by reducing waste, increasing variety, and controlling food quality. As a result of 3D

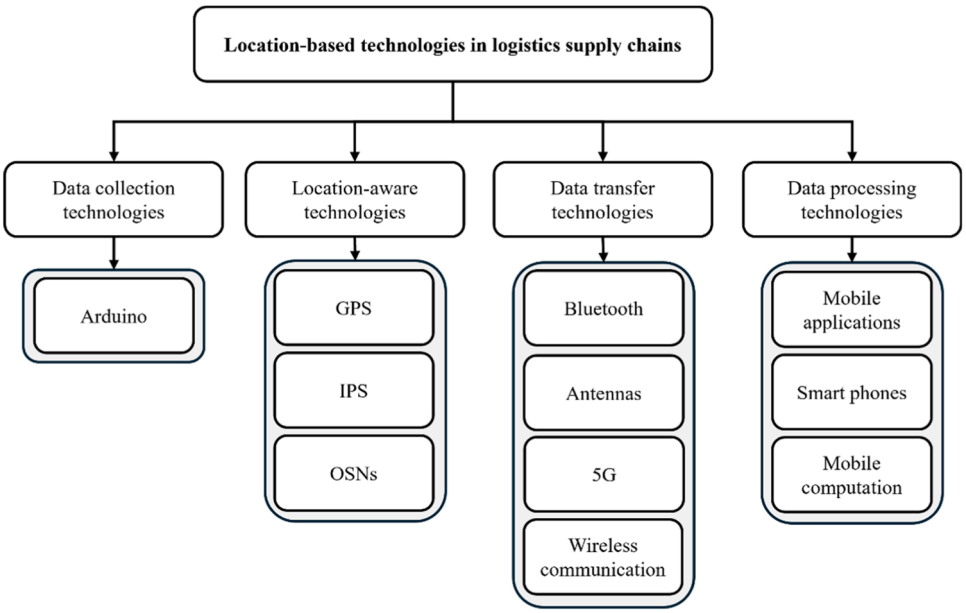

Fig. 12. Enabling technologies of LBSs.

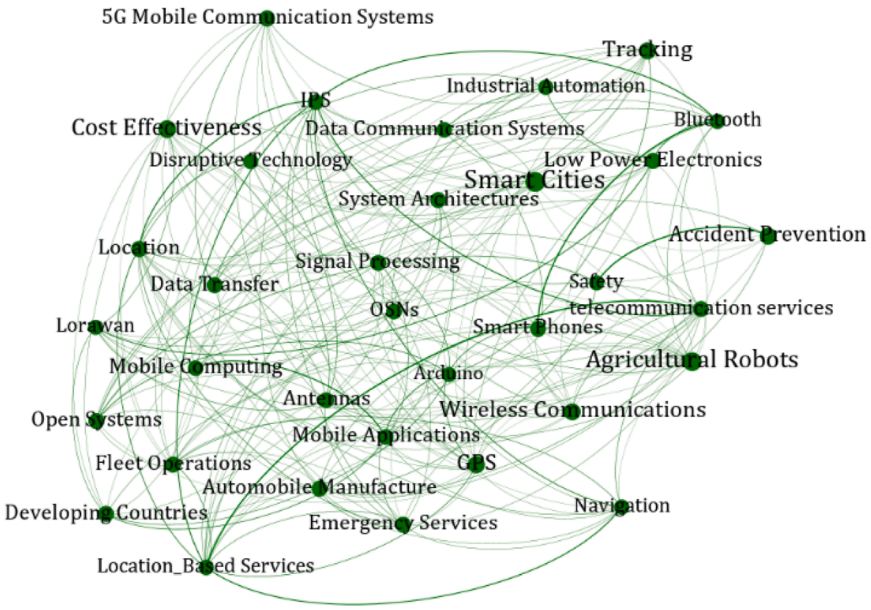

Fig. 13. Wireless communications.

printing, supply chains can design food structures based on customer preferences of color and size [137]. 3D printing eliminates long-distance transportation of products in cold chains by using locally-sourced raw materials in a kitchen. This way, food becomes less susceptible to pathogens, and security increases. 3D printing could be a key technology in pandemic prevention because of the improving food security, especially in the case of the COVID-19 virus [151,91] pandemic. Specifically, C4_3 addressed using 3D printing technology to improve food security, particularly during pandemics such as COVID-19.

4.1.4.4. Digital food storage (C4_4): Blue Sub-theme. Improving food storage in supply chains is one way to reduce food waste [121,130]. Digitalization is improving different parts of the supply chain, such as food storage, so we can reach our food waste reduction goals [6]. Digital platforms, like websites, can store perishable agri-food products, such as fruits, reducing food waste associated with supply chains. Food digital storage for waste management is described briefly in C4_4.

4.1.5. Smart grid challenges (C5)

C5, with 68 nodes and 733 links, describes the challenges of the smart grid in supply chains. After the re-community detection process, the cyber-security area is divided into five main sections, as indicated in Fig. 15.

4.1.5.1. Smart power grids (C5_1): Red Sub-theme. With a smart grid, three components are integrated into existing power grids: modern sensing technologies, advanced communications, and control methodologies [113]. Smart grids utilize automated control and communication techniques to enhance the efficiency and reliability of electric power transmission networks [65]. In a smart grid, electricity is produced by different electricity producers using different energy sources. The generated electricity is then transmitted and distributed to the end users.

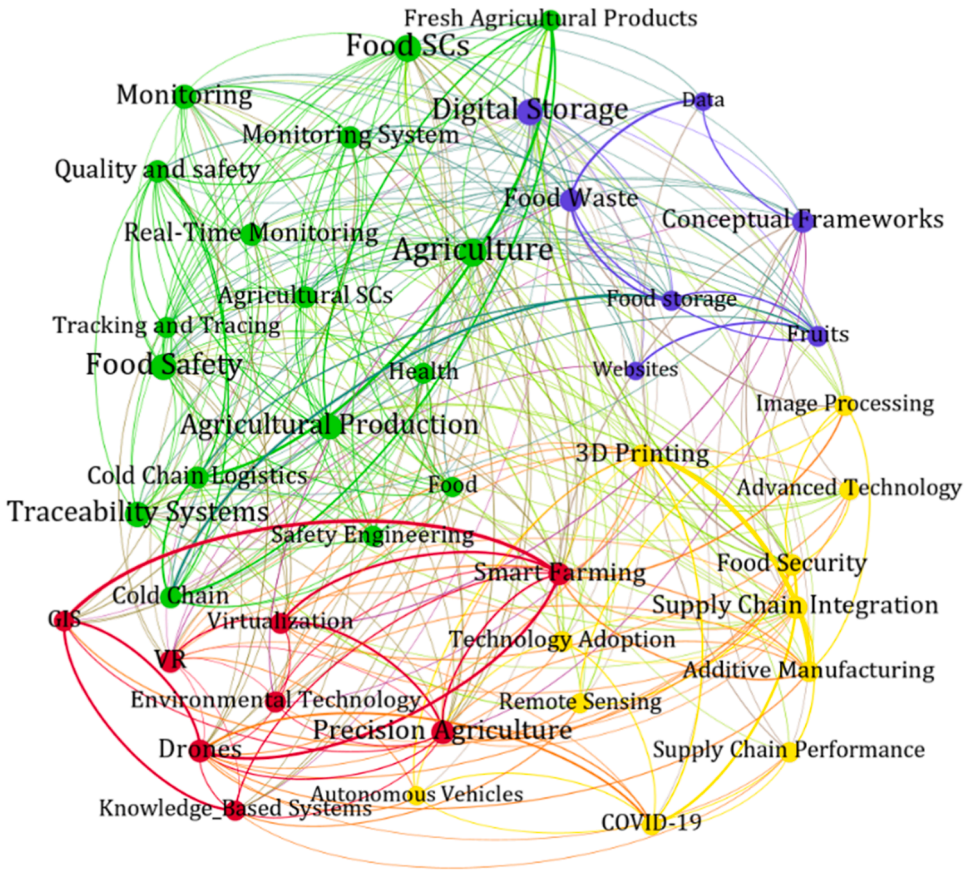

Fig. 14. Agri-food supply chains.

The concept of intelligence applies to load balancing and smart electricity generation in a smart grid concept [102]. C5_1 introduces smart power grids in intelligent electric power transmission for logistical and supply chain sectors, including intelligent vehicle highway systems (IVHS). Electric power transmission in logistics and supply chains is analyzed and controlled by analyzing the data provided by SCADA (Supervisory control and data acquisition) systems.

4.1.5.2. Smart Grids' Vulnerabilities (C5_2): Blue Sub-community. As a type of CPS, smart grids integrate intelligent devices into a communicable setting to make power systems easier to control. However, this integration can pose severe cyber threats, which also present vulnerabilities for smart grids [64]. C5_2 explains the security challenges of DSCs under launching smart power grids. Malware, cyber-attacks such as Denial-of-service attacks (DoS), computer crime, and hardware security are the main components of security issues. Also, anomaly detection and intrusion detection are two standard methodologies aiming to improve the supply chains' smart grid security indicated in the presented sub-graph.

4.1.5.3. Authentication (C5_3): Orange Sub-community. Smart Grid's success depends mainly on how well it can defend itself against remote network-based attacks. The first and most effective defense against attacks is user authentication for smart grid access [54]. Authentication and authentication protocols are the main concepts of C5_3 introducing anti-counterfeiting methods for improving the security issues of smart grids.

4.1.5.4. Supply chain risk management (SCRM) (C5_4): Yellow Sub-community. Cyber-security risk assessment is critical for businesses, which aims to identify vulnerabilities and threats of smart grids. Smart grid security requirements and security controls should be specified using the risks assessed in the risk assessment. Hecht et al., [73]. C5_4 concentrates on implementing risk assessment approaches for identifying cyber-security risks of smart grids in supply chains. The risks associated with supply chains can be managed and controlled through a successful risk assessment.

4.1.5.5. Blockchain-based smart grids (C5_5): Green Sub-theme. As a decentralized transaction and data management technology, Blockchain is one of the Distributed Ledger Technologies (DLTs), working based on peer-to-peer (P2P) systems, where transactions are executed in a distributed environment. Blockchains' ability to provide secure data exchange and management conditions has led many to launch blockchain-based systems [145]. Smart grids have been moving towards blockchain-based smart grids that are more secure and protect themselves from attacks and vulnerabilities [106,123]. C5_2's main focus is on blockchain-based smart grids, improving data privacy, and making supply chains and logistics more secure.

4.1.6. Industry 4.0 (C6)

Industry 4.0 is mainly composed of IoT and CPS, as previously mentioned. C6 with 95 nodes and 1861 edges explains Industry 4.0-related concepts. After the re-community detection phase, seven themes are revealed, as shown in Fig. 16.

4.1.6.1. DSCs (C6_1): Purple Sub-them. Emerging technologies have brought competitive advantages to supply chains, introducing a new concept called DSC, which has resulted in cost reductions [95], product development [33] and performance improvements [92]. C6_1 explains the future directions of supply chains that incorporate emerging technologies and digitalization concepts as competitive advantages. Based

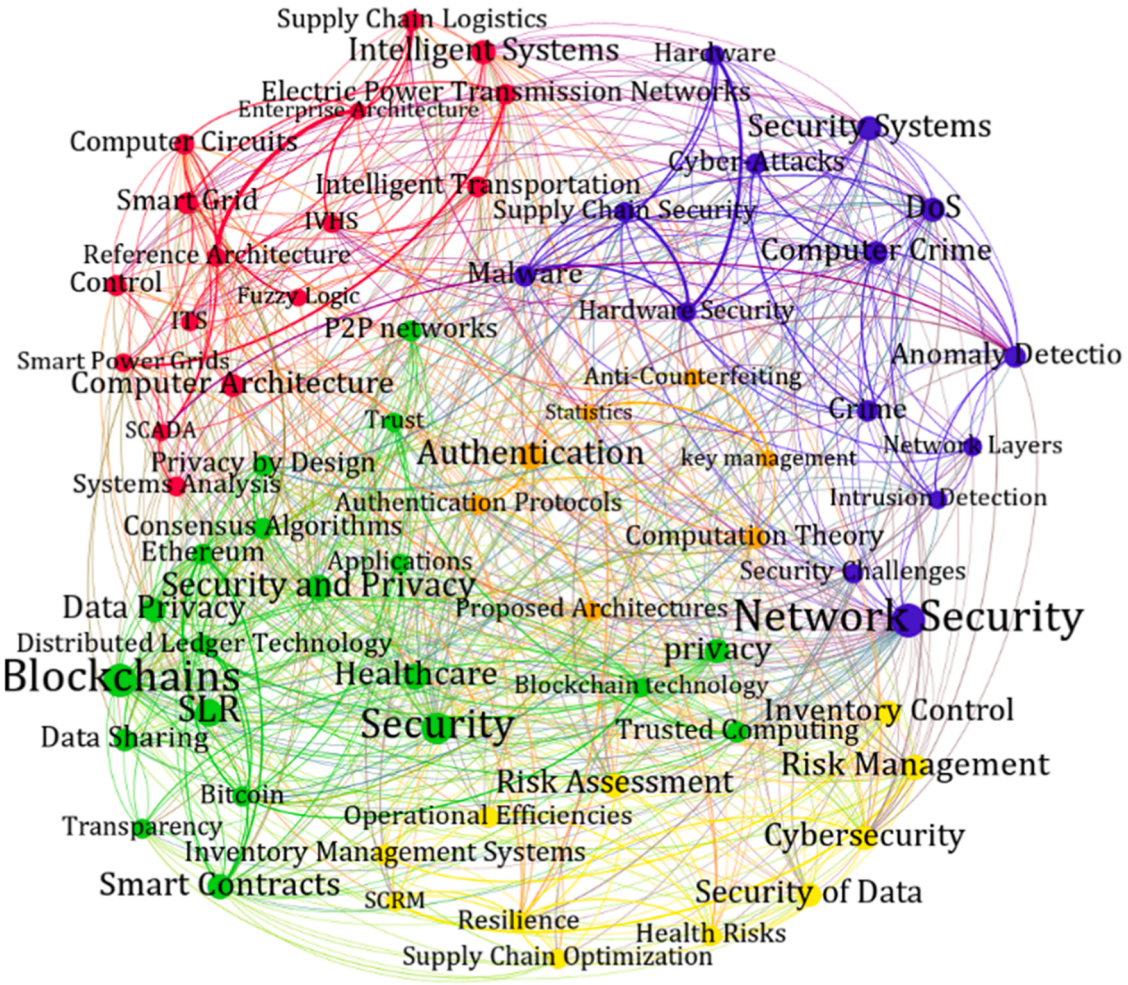

Fig. 15. Cyber-security.

on the provided sub-theme, digitalization is considered the primary purpose of future supply chains.

4.1.6.2. Smart manufacturing (C6_2): Green Sub-theme. Smart manufacturing involves five main concepts, namely: *a) smart designing:* As part of a smart design process, end users can work with manufacturers to develop their desired products; *b) smart monitoring:* with smart monitoring, the status of manufacturing systems can be accessed remotely, and desirable data can be collected and analyzed for specific purposes; *c) smart machining:* it refers primarily to the internet of things-based manufacturing, where automated systems are integrated into a networked system; *d) smart controlling:* is related to managing smart machines involved in the manufacturing process; *e) smart scheduling:* this refers to the planning of the smart manufacturing processes [111,61]. C6_2 focuses on smart manufacturing processes. Nowadays, manufacturing companies apply CPSs and embedded systems to direct manufacturing processes towards smartness. Accordingly, all of the involved implementations of product life-cycle (PLC), such as designing, planning, and controlling, can be executed by smart manufacturing systems.

4.1.6.3. Cloud manufacturing (C6_3): Yellow Sub-theme. Cloud manufacturing combines cloud computing, IoT, and service-oriented technologies to create a collaborative environment between enterprises, even across long distances. Cloud manufacturing has changed how industrial resources are used via internet-based services [29]. Cloud manufacturing is a technology that virtualizes the manufacturing resources to provide ubiquitous demand meeting [101]. C6_3 refers to an approach involving cloud manufacturing in supply chains, where certain services, such as distributed computer systems, are accessible from the cloud.

4.1.6.4. Augmented reality (AR) (C6_4): Red Sub-theme. The goal of AR is to "augment" the physical world with virtual objects. A concept of augmented reality is the overlaying of digital information of physical objects (e.g., geographical places) over computer-created (virtual) settings to improve customer experiences [25]. C6_4 represents the augmented reality technology that uses virtual objects to augment the physical supply chain.

4.1.6.5. Digital twins (DT) (C6_5): Orange Sub-theme. C6_5 refers to the DT technology. A digital twin [12] is a virtual representation of a physical object [63], simulating the real world through collected data from smart devices and sensors. Using mathematical modeling and other practical approaches, DT-based environments can optimize physical environments [109,38]. Although IoT and CPSs enable supply chains and logistics to collect and integrate sensors' data, digital twins capture the collected data and create a copy of the considered supply chain or logistics. Digital twins aid managers trace the system's current situation and enable them to predict the future state of the systems in a simulated environment.

4.1.6.6. Enterprise resource planning and decision support systems (C6_6): Light-blue Sub-theme. Enterprise Resource Planning (ERP) is one of the

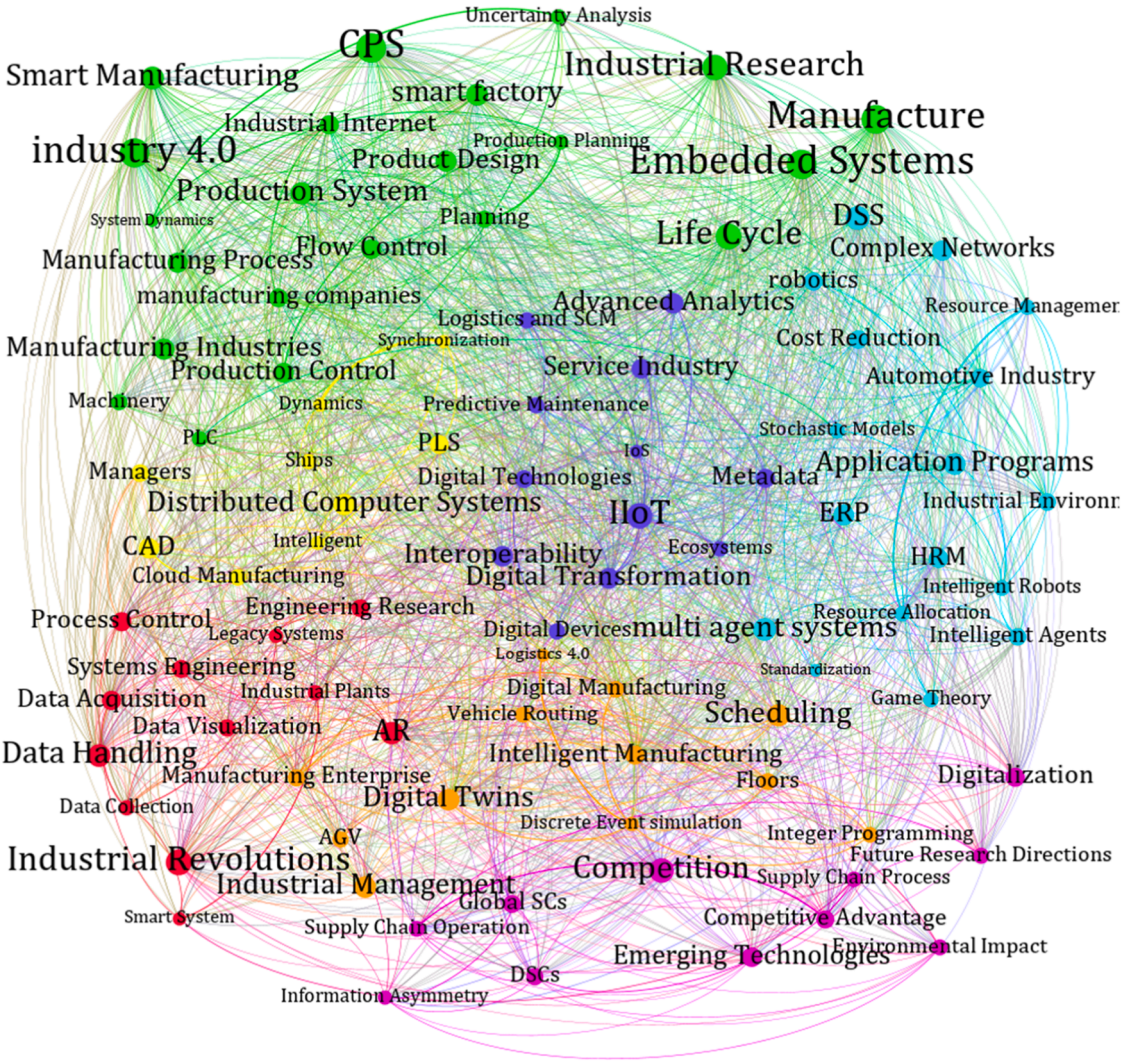

Fig. 16. Industry 4.0.

main crucial elements of organizations and supply chains aiming for efficient resource management and allocation processes. With the advent of innovative technologies, intelligent ERPs emerged, working based on intelligent agents such as intelligent robots providing autonomous cooperation. Decision Support Systems (DSS) is another technology improving the ERPs' performance through analytical tasks.

4.1.6.7. IIoT (C6_7): Dark blue Sub-theme. The IIoT, as a concept of Industry 4.0, is limited to applications of IoT in industries for different goals, such as improving production processes [129]. In C6_7, the IIoT is primarily used to provide industrial services in logistics and supply chain management. Using IIoT technologies, a wealth of data is generated in supply chains and logistics, which can be analyzed to improve managerial tasks.

4.1.7. Intelligent computing (C7)

Evolutionary algorithms as a type of intelligent computing are applied in SCM problems such as energy conservation and sustainability issues. Due to the NP-hard nature of evolutionary algorithms such as Ant-Colony Optimization (ACO) and Particle Swarm Optimization (PSO), novel technologies such as fog computing and edge computing are utilized to increase the performance and speed of the algorithms. This community can be found in Fig. 17.

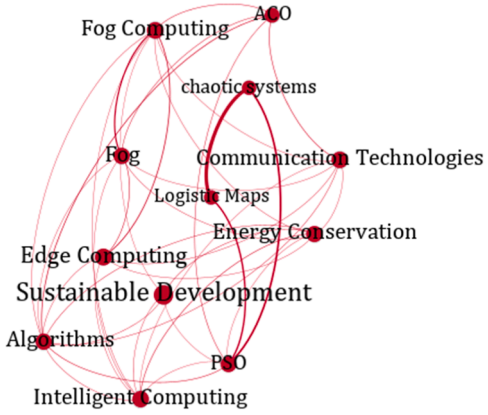

Fig. 17. Fog-based intelligent computing.

4.1.8. Marketing (C8)

C8, with 41 nodes and 376 links, explains the marketing aspect of the considered domain. After running community detection on this theme, four main sub-themes are disclosed, as shown in Fig. 18.

4.1.8.1. Behavioral analysis (C8_1): Orange Sub-theme. Big data enables

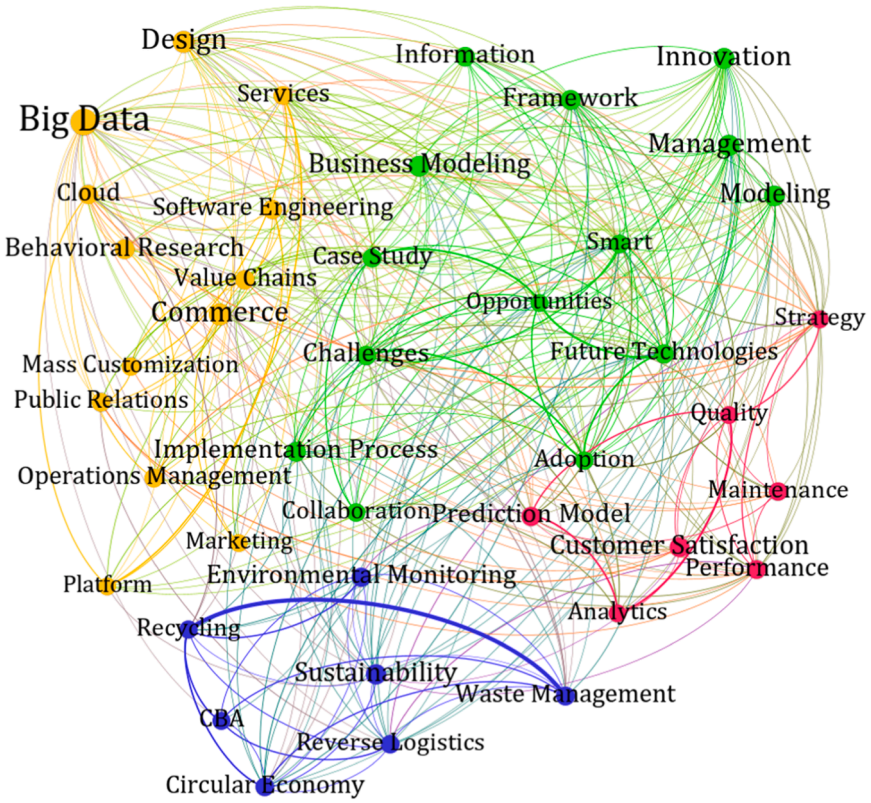

Fig. 18. Feasibility.

marketers and public relations to analyze customers' behavioral characteristics for designing appropriate value chains and services, leading to better operations management and mass customization.

4.1.8.2. Adoption (C8_2): Green Sub-theme. C8_2 mainly focuses on adopting IoT, CPSs, and future technologies in supply chains in which challenges and opportunities are surveyed for implementation processes. Business modeling is one of the effective methods for meeting such an aim.

4.1.8.3. Reverse logistics (C8_3): Blue Sub-theme. Reverse logistics refers to all processes of re-using, re-manufacturing, recycling products, and improving the circular economy of the entire supply chain. Cost and Benefit Analysis (CBA) is one of the exciting investigations executed by researchers in IoT and CPSs-based reverse logistics.

4.1.8.4. Customer satisfaction (C8_4): Red Sub-theme. The quality and performance of the IoT and CPS-based supply chains and logistics are the main factors influencing customer satisfaction. Besides, analytical tools such as prediction models provide relevant insights about customers, guiding managers to select the best strategies to improve customer satisfaction.

4.1.9. Data science (C9)

A massive amount of generated data by embedded systems and sensors of IoT and CPSs, provide conditions for data analysis processes. Data mining, ML, AI domains utilizing different algorithms such as deep learning, neural networks, Support Vector Machine (SVM), logistic regression, Support Vector Regression (SVR), and Random Forest for various goals such as pattern recognition, classification, feature extraction, and forecasting. This theme can be seen in Fig. 19.

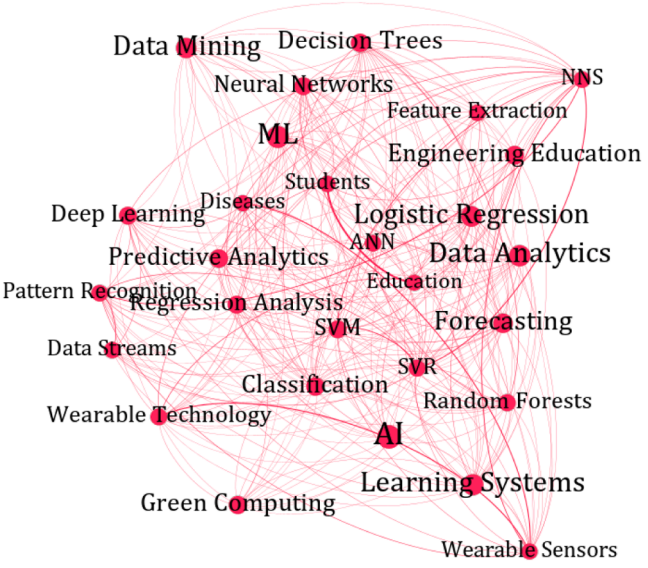

Fig. 19. Data Science.

4.2. Strategic diagram of IoT and CPS-based supply chains and logistics

Although co-word analysis gives an appropriate overview of the investigated topics by disclosing the involved clusters, it cannot plot the status of found communities or clusters. Co-word analysis without employing a strategic diagram would not be beneficial, and one cannot extract practical implications. As a result, a strategic diagram is employed in this research to determine the status of each theme. As explained in Subsection 3.4.2, the strategic diagram works based on the centrality and density measures calculated based on the co-word graph.

The centrality and density of each theme are computed and indicated in Table 4. Overall, the strategic diagram of the revealed communities is indicated in Fig. 20. As shown in Fig. 20, QI is empty. QII (Ivory Tower) comprises three themes of "Intelligent Computing C7", "Marketing C8", and "data science C9" which are well structured internally but without solid external linkage.

Connecting these themes with the other eight communities extends and moves them to the QI area. QIII, with four themes, has the greatest number of areas, namely "Project Management C1", "Wireless Communication C3", "Agri-Food Supply Chains C4", and "Smart Grid Challenges C5". Indeed, a tremendous amount of studies and investigations are in their initial stages. C1 and C3 are closer to QII, while C4 and C5 are closer to QIV. Accordingly, C1 and C3 need more investigations, and C4 and C5 require more combinations with other themes for moving to QIV.

Generally speaking, QIV includes two central themes: "enabling technologies" and "Industry 4.0". Because of the high-density measure of the themes, little effort on these themes transfer theme into QI.

5. Discussion and insights

Regarding the strategic diagram, the ideal point for each theme lies in QI, in which themes are developed enough and connected to other related themes intensely. In other words, the closer the strategic diagram's top-right corner, the more mature theme. The nature of well-developed and external-connected themes of QI has comprehensive effects on other scopes, guiding them to executing novel research. Hereupon, they play a motor role for other domains and are known by the 'mainstream' metaphor. Fig. 20 reveals an absence of themes within Quadrant I, which may suggest that the maturity of IoT and CPS-based supply chains is still in development. This could potentially lead to an intensified focus among researchers on this topic, thereby increasing the volume of related publications. It is recommended that themes from other quadrants be directed towards Quadrant I to enhance their level of maturity.

5.1. QII: Ivory tower

Many researchers have well investigated and developed some themes but without enough combination with other related domains, which gives rise to staying on QII, such as C7, C8, and C9. For this reason, themes of QII are called 'developed but isolated themes'. Indeed, despite enough growth in academic aspects of QII, they do not have much usability in real-world conditions. *Ivory Tower* is an appropriate metaphor applied for QII, referring to 'science without application'.

C7, or Intelligent Computing, has theoretical and technical complexity, preventing researchers of different fields from investigating it. Also, examining such technologies needs to be implemented in many cases that reduce the research speed and make it complicated. Accordingly, researchers in other fields have been less interested in studying this area. The initialization of online fog-based services with easy-to-use ability enables researchers to investigate similar domains effectively.

C8 or marketing is another isolated field that could grow internally well, concentrating on the consumers' sides. Overall, marketing could grow internally in different themes, i.e., customers' behavioral analysis, technology adoption, customer satisfaction, and reverse logistics, while it could not penetrate other domains. Indeed, a wide range of domains affects companies' choices to apply IoT and CPSs in their supply chains which were neglected in the C8, making this field more isolated. For instance, the two main concerns of supply chains are cyber-security and health fields that directly affect technology adoption in market research.

According to C9, data science techniques are applied to the generated data by CPSs and IoT technologies for analyzing the supply chain sections. Although data analytic tools were utilized chiefly, they were employed in a few applications because of the low amount of accessible data in this field. The security issue is one of the main reasons preventing data sharing in this field. Launching simulation-based systems can provide related data for various analytical applications in different domains.

Table 4
Re-community detection results.

Id	Cluster	Color	No. Nodes	No. Edges	Modularity	No. Sub-themes
C1	Project Management	Pink	11	21	-	1
C2	Enabling Technologies	Light-green	138	3803	0.250	7
C3	LBSs	Dark green	34	214	-	1
C4	Agri-Food Supply Chains	Dark blue	43	347	0.339	4
C5	Smart Grid Challenges	Light blue	68	733	0.416	5
C6	Industry 4.0	yellow	95	1861	0.260	7
C7	Intelligent Computing	Dark Red	12	37	-	1
C8	Marketing	Orange	41	376	0.309	4
C9	Data Science	Light red	28	292	-	1
Total		-	**470**	**27423**	**0.252**	**9**

5.2. QIII: Emerging or declining themes

QIII is the opposite of QI, where both maturity criteria (development and connection) are low. Overall, themes of this quadrant lie in two sections: a) emerging themes and b) declining themes. Due to the newness of the areas of this section, many irregularities can be seen in it, indicating an undeveloped sphere. Therefore, the '*Chaos/undeveloped*' metaphor is used for QII. Because of the short life of IoT and CPS-based supply chains, themes of QIII are considered emerging. C1, C3, C4, and C5 are the components of QIII.

Since IoT and CPS-based supply chains are new, particularly in the real world, researchers faced a lack of standard models for implementation. Accordingly, an endeavor was made to develop simulation-based methods for virtualizing different states of technology implementation. Using such methods speeds up research processes without any time-consuming implementation.

C3 or Location-based Services has mainly concentrated on applicable hardware of IoT and CPSs in supply chains. IoT and CPSs have been applied for a short time in supply chains, resulting in low testing of related technologies. Accordingly, researchers may investigate and compare the more comprehensive range of hardware devices in supply chains.

Due to the many benefits of IoT and CPS technology in perishable supply chains, some of them, such as agri-food supply chains (C4), could be emerged as an independent clusters in the studied network. The high risk of agri-food supply chains for human health, particularly in the COVID-19 crisis, is another reason that has forced researchers to investigate IoT and CPS technologies in them. Most applications of IoT and CPS-based technologies in agri-food supply chains refer to real-time tracking tasks in which product locations and conditions (such as humidity and temperature) are traced everywhere and every time. Connecting collected data from tracking systems with analytical tools can add other beneficial tasks to supply chains, such as predicting tools. A prediction-based system enables supply chains to predict critical factors such as delays, spoiling time, supply, and demands.

Although cyber-security is one of the main problems of IoT and CPS-based systems, it is in its infancy due to its new applications in supply chains. One of the leading solutions for cyber-security problems is implementing Blockchain technology utilizing distributed computing, resulting in a safe IoT and CPS-based supply chain. It can be predicted that applying Blockchain technologies in IoT and CPS-based supply chains will soon change into a hot topic domain.

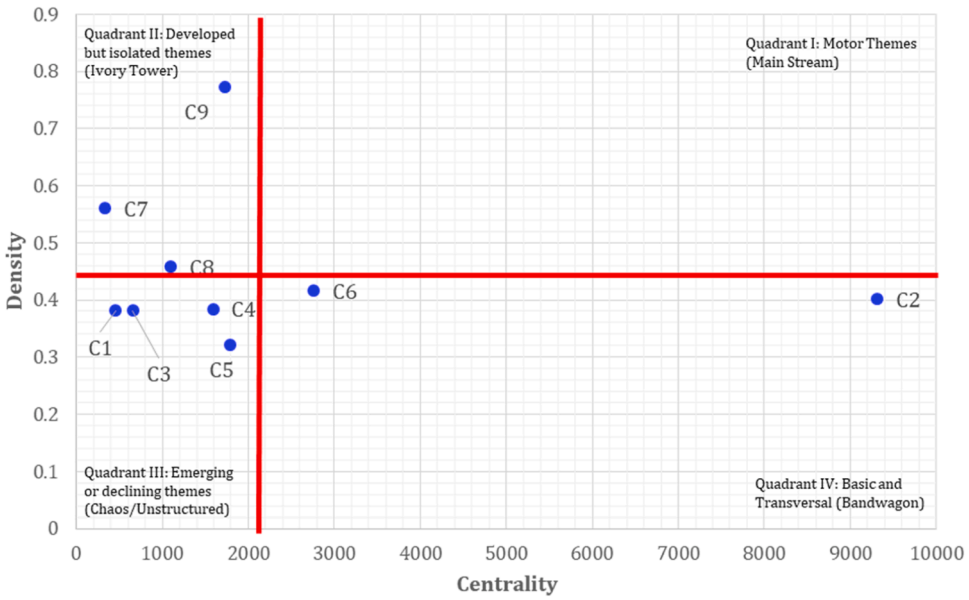

Fig. 20. Strategic diagram.

Table 5
The strategic diagram measurement.

Q	ID	Top Keywords	Centrality	Density
QI: Motor	-	-	-	-
QII: Ivory Towers	C7	Intelligent computing, fog computing, edge computing, ACO, PSO	336	0.5606
	C8	Big data, behavioral research, CBA, customer satisfaction, value chains	1101	0.4585
	C9	Data mining, ML, AI, pattern recognition, deep learning, learning systems	1728	0.7725
QIII: Emerging/ Declining	C1	Simulation model, computer simulation, software, standards	457	0.3818
	C3	Arduino, GPS, antennas, LoRaWAN, IPS, Bluetooth	655	0.3815
	C4	Agricultural products, food SCs, cold chain logistics, food waste, precision agriculture	1591	0.3843
	C5	Cyber-attacks, blockchain, computer crime, security challenges, authentication, privacy, network security	1786	0.3218
QIV: Bandwagon	C2	IoT, RFID, WSN, Zigbee, LIS, WoT, semantic web, M2M, CC, ICT	9315	0.4023
	C6	Industry 4.0, IIoT, AR, ERP, embedded systems, digital twins, smart manufacturing, DSC	2760	0.4168

5.3. QIV: Basic and transversal

The QIV theme refers to basic themes that cannot be developed fully theoretically but are applied across many applications. Though a theme may be developed well, only its basic knowledge is applied to the field under investigation. Perhaps industry 4.0 is well developed, but in the context of IoT- and CPS-based supply chains, only the basics of the concept are used. The IoT- and CPS-based supply chains are not well-developed in industry 4.0, but their fundamental concepts have considerable applications. They are considered '*bandwagons*', employed by other domains. C2 and C6 are categorized in QIV.

The concept of C2 refers to the enabling technologies of IoT and CPS implemented in several aspects of supply chains. Technology-enabled supply chains and logistics are considered the foundation of the IoT and CPS. While related technologies are advancing rapidly, learning about and implementing them in supply chains takes time. Therefore, the investigation of new IoT and CPS technologies from the DSC perspective has remained in the early stages. Research and Development (R&D) organizations that survey novel technologies used in supply chains may encourage researchers to include novel technologies in different aspects of DSCs.

The focus of C6 or Industry 4.0 is heavily on the manufacturing aspects of IoT- and CPS-based supply chains connected to the IIoT. C6 is interpreted very similarly to C2, in which a wide range of related technologies has been investigated in manufacturing supply chains. Moreover, research on the technological aspects of each technology was neglected, and utilizing related technical experts can give a better theoretical overview to audiences.

In the dynamic domain of digital supply chains, it is imperative to bridge the gap between theoretical advancements and practical applications. The domains of Intelligent Computing, Marketing, and Data Science have reached a level of theoretical maturity that positions them as prime candidates for pioneering application-driven research. These areas are ripe for innovation, offering a fertile ground for the development of cutting-edge features that can revolutionize data-driven technologies. Conversely, the domains of enabling technologies and Industry 4.0 have seen widespread practical application across various sectors within digital supply chains. Despite this, there exists a significant opportunity to bolster their theoretical foundations. By deepening the theoretical understanding, we can unlock new dimensions of these technologies and enhance their efficacy and scope of application. Furthermore, emerging themes such as Project Management, Location-Based Services (LBSs), Agri-Food Supply Chains, and Smart Grid Challenges present a dual frontier for development. These areas require both theoretical expansion and empirical exploration to uncover novel applications. It is essential to foster a symbiotic relationship between theory and practice, ensuring that each informs and enriches the other.

6. Conclusions, limitations, and future research

This paper concentrated on disclosing the main themes of CPS and IoT-based supply chains and logistics. A co-word analysis, one of the significant bibliometric methods, was applied to related publications of Scopus and web of science (WoS) databases between 2010 and 2021.

Generally, nine central clusters were revealed, including 1) "Project Management", 2) "Enabling Technologies", 3) "Location-based Services", 4) "Agri-Food Supply Chains", 5) "Smart Grid Challenges", 6) "Industry 4.0", 7) "Intelligent Computing", 8) "Marketing", and 9) "Data Science".

Upon identifying the core themes, we utilized a strategic diagram to assess the present status of each topic within IoT- and CPS-driven supply chains. The analysis revealed an absence of a mainstream theme—a position signifying extensive development and widespread application across various domains. Consequently, the goal for each theme is to achieve enhanced development and increased applicability, propelling it into a mainstream status. Themes C7, C8, and C9 reside in the Ivory Tower quadrant, indicating they are underdeveloped. To transition into mainstream topics, they need to be integrated with other relevant themes. Themes C1, C3, C4, and C5 are positioned in QIII, signifying they are either emerging or declining themes that are peripheral and not fully developed. These themes exhibit numerous research gaps, necessitating focused attention for comprehensive internal and external exploration. It's imperative that each theme undergoes thorough investigation and interdisciplinary collaboration. Themes C2 and C6, located in the bandwagon quadrant, are more central yet less developed. Intensified internal research efforts could propel them towards the motor quadrant, which is indicative of well-developed and widely applied themes.

Although this paper investigated CPS- and IoT-based supply chains successfully, it dealt with some critical limitations, which could be considered as future streams, as follows:

6.1. Time-consuming

Co-word matrix calculation is time-consuming, mainly when the number of investigated publications and keywords is high. Utilizing parallel computing and big data approaches can improve this issue effectively.

6.2. Prediction model

The proposed co-word analysis describes the current situation of the publications without prediction ability. Combining methods such as link prediction with co-word analysis gives a comprehensive overview of future research.

6.3. Web mining

The rise of blogs and OSNs has significantly facilitated the accessibility of data [18,88]. The present research was conducted on academic databases while there is a massive amount of data about CPS- and IoT-based supply chains on websites and OSNs. Besides, due to the relatively long review, published papers' contents in WoS and Scopus databases are older than websites, blogs, and OSNs. Accordingly, implementing the co-word analysis on these sources may extract another aspect of knowledge. It must be considered that this process needs an appropriate keyword extraction algorithm for running co-word analysis on web-based data because web-based datasets are unstructured and have no keyword section, such as publications of WoS and Scopus.

CRediT authorship contribution statement

Aseem Kinra: Supervision, Project administration, Methodology, Investigation, Conceptualization. **Vahid Kayvanfar:** Writing – original draft, Supervision, Project administration, Methodology, Conceptualization. **Hamed Baziyad:** Writing – original draft, Visualization, Validation, Software, Methodology, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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